INCH-POUND
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MILITARY SPECIFICATION

SWITCH, PROXIMITY, SOLID-STATE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope This specification establishes the performance, test, manufacture, and acceptance requirements for shipboard solid-state proximity switches The switches use an inductive sensor for noncontact position sensing of metallic objects. The switches transfer power between a power source and its loads when the switch is actuated.

1 2 Classification Solid-state proximity switches shall be classified using the following classifications as specified (see 6 2 1)

Class I (XX) - Type of input voltage.
Class II (Y Y) - Maximum sensing distance
Class III (ZZZZ) - Output states.

1.2.1 Switch designations Switches shall be designated according to type of impact voltage (alternating current (ac) or direct current (dc)), maximum sensing distance in inches, and output states (NO for normally open, NC for normally closed, and NONC for both normally opened and normally closed), as follows (see 6.2.1).

Ac - 0 5 - NO
Ac - 1 5 - NO
Ac - 0 5 - NC
Ac - 1 5 - NC
Dc - 0 5 - NONC
Dc - 1 5 - NONC

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated
MIL-S-22698	- Steel Plate and Shapes, Weldable Ordinary Strength and Higher Strength Hull Structural
MIL-F-24385	- Fire Extinguishing Agent, Aqueous Film-Forming Foam (AFFF) Liquid Concentrate, for Fresh and Sea Water.
MIL-S-28786	- Switches, Packaging of.

STANDARDS

MILITARY

MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-704	- Aircraft Electric Power Characteristics.
MIL-STD-1399, Section 300	- Interface Standard for Shipboard Systems Electric Power, Alternating Current (Metric)

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Switch characteristics.

3.2.1 Ac switches, electrical characteristics. Ac switches shall be two-wire devices and shall operate in series with the load Each switch shall have one NO or NC solid-state output.

3.2.1.1 Operating voltage. The load and switches shall operate across a single-phase power source of 65 to 130 volts alternating current (Vac), 60-hertz (Hz) Operation for 115 Vac shall be in accordance with MIL-STD-1399, section 300, type V (see 4.5.1).

3 2.1 2 Voltage drop The voltage drop across each switch, during the activated (ON) state, shall not exceed 4.0 Vac root mean square (rms) at rated load current (see 4 5.1)

3.2 1 3 Leakage current. The leakage current through each switch, during the deactivated (OFF) state, shall not exceed 2 0 milliamperes (mA) rms (see 4.5 1).

3.2.1 4 Load ratings The load rating of the switches shall operate as specified (see 4 5.1) The load ratings of the switches shall be as specified in 3 2 1 4 1 through 3 2 1 4 4

3 2.1 4.1 Resistive Switches shall have a resistive rating of 1 25 amperes rms.

3.2 1.4.2 Inductive. Switches shall operate inductive loads with a power factor between 1 and 0.35. As a minimum, switches shall have a make-current rating of 10 0 amperes rms for three cycles of the specified operating voltage and a break current rating of 1.25 amperes rms

3.2 1.4.3 Minimum load Switches shall operate with a minimum load of 15 mA rms

3 2 1 4.4 Overload. Absolute overload and short circuit protection are desired As a minimum, switches shall be protected from continuous overloads up to 6 0 amperes rms. Interruption of the operating voltage shall be required to restore normal operation of the switch after an overload has been detected.

3.2 1.5 Multiple switch operation Multiple switch operation shall be as specified (see 4 5.1)

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3.2.1.5.1 Series connection. A minimum of six switches shall operate in series. The total switch voltage drop at the load shall not exceed 4.0 Vac rms times the number of switches in series.

3.2.1.5.2 Parallel connection. A minimum of six switches shall operate in parallel. The total switch leakage current through the load shall not exceed 2.0 mA rms times the number of switches in parallel.

3.2.1.6 Transients. Switches shall continue to operate without degradation when the operating power or output is subjected to voltage and frequency transients in accordance with MIL-STD-1399, section 300 for 115-Vac, type I power (see 4.5.1). The spike voltage test amplitude shall be 2500 volts The spike waveform shall be in accordance with MIL-STD-1399, section 300. Application of a deviation in voltage over a deviation in temperature (dV/dT) of 1000 volts per microsecond across the output contact leads at the end of the supplied cable shall not result in false closure or opening of the output.

3.2.1.7 Dielectric withstand. The switches shall withstand a plus or minus 250 volts direct current (Vdc) potential between switch case and each individual input and output connection for 60 seconds with less than 0 75 mA rms leakage current flow (see 4.5.8).

3 2.1.8 Insulation resistance. Insulation resistance between the switch case and each individual input or output lead or operational power lead shall be greater than 10 megohms. Measurement shall be in accordance with MIL-STD-202, method 302, test condition A for a duration of 2 minutes The test may be terminated before the end of 2 minutes if the measured insulation resistance is greater than 10 megohms and is steady or increasing.

3.2.1.9 Ac wire colors. Wire colors shall be as follows:

NO switches:	NC switches:
Black - Input	Black - Input
White - Output	White - Output

3.2.2 Dc switches, electrical characteristics. Dc switches shall be four-wire devices and shall operate as voltage sources. Switches shall be line-powered and shall have solid-state outputs. Each switch shall have one NO and one NC output in a complementary configuration. The total power drawn from the line shall not exceed the sum of 50 mA plus the output load current.

3.2.2.1 Operating voltage. Dc switches shall operate with any voltage between 10 and 30 Vdc (see 4.5.1).

3.2.2.2 Voltage drop. The voltage drop across each switch, during the activated (ON) state, shall not exceed 1.5 Vdc at rated load current (see 4 5 1)

3.2.2.3 Leakage current. Leakage current from each output in the open (OFF) state shall not exceed 300 microamperes with 35 Vdc applied to the switch as specified (see 4.5.1).

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3.2.2 4 Load ratings. The switch shall operate as specified with the load ratings of the switch as specified in 3.2.2.4.1 through 3.2.2.4.3 (see 4.5.1).

3.2.2 4.1 Maximum current The switches shall continuously supply 250 mA to resistive and inductive loads and 100 mA into lamp loads The inductive load shall have a maximum decay time of 100 milliseconds when the load is interrupted This decay shall be measured from the 90 percent level to the 10 percent level.

3 2.2.4 2 Minimum load. The switches shall supply any load current from maximum down to zero.

3.2.2.4.3 Overload. The switches shall be protected from damage due to any overload or direct short circuit of the output. Interruption of the operating voltage shall be required to restore normal operation of the switch after an overload has been detected

3.2.2.5 Reverse polarity protection Switches shall be immune to damage from reversal of the input power or output leads (see 4.5.1).

3 2.2 6 Transients. Switches shall continue to operate without degradation when the input power and outputs are subjected to voltage transients in accordance with MIL-STD-704 for 28-volt systems (see 4 5 1)

3.2 2.7 Dielectric withstand The switches shall withstand a plus or minus 250 Vdc potential between switch case and each individual input and output connection for 60 seconds with less than 0.75 mA rms leakage current flow. Measurement shall be in accordance with MIL-STD-202, method 301 (see 4.5 8).

3.2.2.8 Insulation resistance. Insulation resistance between the switch case and each individual input or output lead or operational power lead shall be greater than 10 megohms Measurement shall be in accordance with MIL-STD-202, method 302, test condition A for a duration of 2 minutes. The test may be terminated before the end of 2 minutes if the measured insulation resistance is greater than 10 megohms and is steady or increasing

3.2 2 9 Dc cable wiring. Wire colors shall be as follows.

Black - Dc power high (plus lead)
White - NO output
Red - NC output
Green - Dc power low (minus lead)

3.2.3 Operation

3.2.3 1 Sensing distance. The sensing distances for the switches shall be as follows. (see 4.5.1)

(a) For all 1.5-inch switches, the sensing distance in the head-on mode shall be 1 50 plus 0.50 minus 0.12 inches Measurements shall be made using a standard target as specified in 3.2 3.4 1

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(b) For all 0.5-inch switches, the sensing distance shall be 0.5 inches plus or minus 0.05 inches in the head-on mode Measurement shall be made using a standard target as specified in 3.2.3.4.1.

The sensing distance of the switches shall operate as specified (see 4.5.1).

3.2.3.2 Hysteresis. Hysteresis in the head-on mode shall be a minimum of 5 percent of nominal sensing distance and a maximum of 10 percent of nominal sensing distance (see 4.5.1).

3.2.3.3 Repeatability. Repeatability characteristics of the switches shall be as specified in 3.2.3.3.1 and 3.2.3.3.2.

3.2.3.3.1 Repeatability to actuation curve. The sensing distance in the slide-by mode (see figure 1) at any distance from the sensing face, shall not vary more than plus or minus 0.12 inch from the value on the contractor's actuation curve. The repeatability shall be true over the specified voltage and temperature ranges. The repeatability shall also be true at any distance between 0.0 to 1.12 inches for all 1.5-inch switches (see figure 2) and between 0.0 to 0 38 inch for all 0.5-inch switches from the sensing face (see figure 3)

3.2.3.3.2 Actuation curve slope requirements. The slope of the actuation curve shall be as steep as possible (see figures 2 and 3). As a minimum, it shall meet the following for all 1.5-inch switches in the slide-by mode Using the standard target at a range of 0.75 inch above the sensing face, the actuation point shall be located in the slide-by mode while moving the target in a plane parallel to the sensing face. The perpendicular distance shall be measured from the reference axis to the actuation point. While moving the target in a plane parallel to the sensing face at a range between 1.0 and 0.5 inch above the sensing face, the perpendicular distance from the reference axis to the actuation points shall not vary by more than plus or minus 0.25 inch from the established distance at a range of 0.75 inch.

3.2.3.4 Target. A typical target shall be a common grade of ferrous metal in accordance with MIL-S-22698 and shall require no special treatment The target shall not be active in any manner or magnetization. The sensor shall respond to the target bulk material properties and not to any surface finish Target size shall be greater than 3 inches round or square for all 1.5-inch switches, and greater than 2 inches round or square for all 0.5-inch switches Target thickness shall be 0.062 inch minimum, and the thickness shall be uniform

3.2.3.4.1 Standard target. The standard target shall be 4130 steel 0.062 inch thick as specified in MIL-S-22698. For all 1.5-inch switches, the standard target shall be 4.0 inches square. For all 0.5-inch switches the standard target shall be 3 inches square. These standard targets shall be used to demonstrate conformance to this specification. Other target shapes, sizes, and materials may be used in service.

3.2.3.5 Response time. The switches shall change operating state within 50 milliseconds after the target is removed or inserted for any load conditions (see 4.5.1).

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3.2.3.6 Adjacent metals. The switches shall operate as specified herein in the presence of metal against the back of the case or against the side of the case up to the plane of the sensing face. Metal at the level of the sensing face or above shall be at least one-half the sensing face diameter away from the edge of the sensing face (see 4.5.1).

3.2.3.7 Facing metals. The switches shall operate as specified herein in the presence of ferrous metal directly in front of the sensor face as close as three times the sensing distance. For non-ferrous metals, the distance may be as close as two times the sensing distance (see 4.5.1).

3.2.3.8 Adjacent sensors Multiple sensors shall operate as specified herein when placed side by side with the distance between adjacent edges of the switches greater than the sensing distance. Sensors shall operate as specified herein when placed face to face with the distance between sensing faces greater than six times the sensing distance (see 4.5 1)

3.2.3.9 Magnetic attraction. The sensor shall not establish a magnetic field so strong as to attract or hold dust size or larger ferromagnetic materials to the sensor face. Dust size particles shall be standard reagent quality iron powder (100 to 200 mesh) (see 4.5.1).

3.2 3.10 Switch stabilization Switches shall stabilize to normal operation within 500 milliseconds of initial application of operating power. False output transitions shall not occur during power up when the minimum dV/dT of the applied voltage is greater than 200 volts per second (see 4.5 1).

3.2.4 Physical characteristics.

3.2.4 1 Weight The weight of each switch, excluding the cable, shall not exceed 7.0 pounds (see 4.4.3).

3.2 4 2 Dimensions The distance between the mounting bolt hole centerlines shall be 3 8 inches for all 1 5-inch switches and 3 inches for all 0 5-inch switches Other dimensions shall not exceed those maximum dimensions shown on figures 4 and 5, as applicable (see 4 4.3)

3.2.4.3 Cable. The contractor shall supply a resilient, UL-listed cable with each switch. The cable shall be 20 feet in length If shielded cable is selected, the shield shall not be electrically connected to the switch case. The cable shall be selected based on previously specified color coding, adequate electrical characteristics and the following physical properties: weatherability, moisture resistance, cold flexibility, flame retardancy, resistance to mechanical abuse, aqueous film forming foam (AFFF) compatibility and chemical inertness. Conductors shall be stranded. The cable outside diameter shall not exceed 0.5 inch. The cable shall be strain-relieved and sealed into the switch (see 4.4.3). (For cable recommendations, see 6.5 1)

3.2 4.4 Encapsulation The internal electronic circuitry and cable entry shall be encapsulated (see 4.4.3) The contractor shall select an appropriate compound which will enhance the ability of the switch to meet all requirements of this specification (see 6.5 2)

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3.2.5 Reliability. Switches shall have a minimum Mean-Time-Between-Failure (MTBF) of 45,000 hours or 500,000 actuation cycles at full rated load (see 4.5.2).

3.2.6 Maintainability. Switches shall be non-repairable and non-adjustable as a final item.

3.2.7 Environmental factors.

3.2.7.1 Temperature. Switches shall operate within specified accuracy in ambient temperature ranging from minus 20 to plus 70 degrees Celsius (°C) (minus 4 to plus 158 degrees Fahrenheit (°F)) (see 4.5.3).

3.2.7.2 Humidity. Switches shall operate within specified accuracy in 0 to 100 percent humidity (see 4.5.3).

3.2.7.3 Vibration. Switches shall withstand type I vibration in accordance with MIL-STD-167-1. Vibration test frequency shall be from 4 to 33 Hz. Switches shall remain energized and there shall be no electrical or mechanical damage. There shall be no momentary change of output states (see 4.5 4).

3.2.7.4 Shock. Switches shall meet the shock requirements specified in MIL-S-901 for grade A, type A, class I, lightweight equipment Switches shall operate during and after the shock test (see 4.5.5) and no portion of the switch shall break off or otherwise create a hazard to personnel or other components There shall be no momentary change of output states. Following the test, the switches shall pass the dielectric withstand voltage test (see 4.5.8).

3.2.7.5 Salt spray. Switches shall operate after being subjected to the salt spray test specified in MIL-STD-202, method 101D (see 4.5.6).

3.2.7.6 AFFF compatibility. Switches shall be compatible with the AFFF in accordance with MIL-F-24385 (see 4.5.10). Compatibility shall be with both liquid AFFF and AFFF foam.

3.3 Materials. Materials shall be as specified (see 4.4.3).

3.3.1 Case and mounting plate. The contractor shall select an appropriate corrosion-resistant material for the switch case and mounting plate, if used If the material is a non-metal, it shall be resistant to damage from asymmetric mechanical loading. Painting or other surface treatment shall not be required to provide corrosion-resistance. Surface shall not require maintenance (see 6 5.3).

3.3.2 Electrical components. Electrical components shall be in accordance with MIL-E-917 and MIL-STD-454.

3.4 Electromagnetic compatibility. Switches shall conform to the requirements of MIL-STD-461, class A4, surface ships, and shall be capable of operation in a space with a maximum field strength of 100 volts/meter during exposure to continuous wave frequency fields and 200 volts/meter during exposure to pulsed wave frequency fields with the supplied cable length exposed.

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3.5 Identification and marking.

3.5.1 Identification label. The contractor shall provide a label on the back (the side facing the mounting surface) of each switch The label shall not be deteriorated by exposure to salt water, glycol, petroleum products or salt water AFFF solution as specified in MIL-F-24385. As a minimum, the label shall contain the following information.

(a) Identification as a proximity switch
(b) Classification designation.
(c) Manufacturer's code or part number traceable to date of manufacture.
(d) Serial number.
(e) National stock number (NSN).
(f) Wiring schematic for hook-up of end function of wires by color code
(g) Note stating "DO NOT MEGGER".
(h) Manufacturer's name and address

3.5 2 Marking requirements Identification as a proximity switch and the NSN shall be marked on the switch so as to be readily visible when the switch is mounted Marking shall be accomplished by raised or depressed letters not less than 0.02 inch in height or depth.

3.6 Drawings When specified in the contract or order, drawings shall be prepared (see 6.2 2).

4. QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4 1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4 2 Classification of inspection The inspection requirements specified herein are classified as follows:

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(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

(1) Production inspection (group A, see 4.4.3).
(2) Production control inspection (group B, see 4.4.4).
(3) Environmental inspection (group C, see 4.4.5).

4.3 First article inspections. The items designated for first article inspection (see 4.3.1) shall be subjected to all examinations and tests necessary to determine compliance with this specification. Tests shall be as specified in table I.

TABLE I. Examination and tests.

	Requirement	Test method	First article	Quality conformance inspection		
				Group A	Group B	Group C
Operating voltage	3.2.1.1 and 3.2.2.1	4.5.1	X	X		
Voltage drop	3.2.1.2 and 3.2.2.2	4.5.1	X	X		
Current leakage	3.2.1.3 and 3.2.2.3	4.5.1	X	X		
Load ratings	3.2.1.4 and 3.2.2.4	4.5.1	X	X[1/]		
Sensing distance	3.2.3.1	4.5.1	X	X		
Multiple switch operation	3.2.1.5	4.5.1	X		X	
Transients	3.2.1.6 and 3.2.2.6	4.5.1	X		X	
Reverse polarity protection	3.2.2.5	4.5.1	X		X	
Hysteresis	3.2.3.2	4.5.1	X	X		
Response time	3.2.3.5	4.5 1	X		X	
Repeatability	3.2.3.3	4.5.1	X		X	
Adjacent metals	3.2.3.6	4.5.1	X		X	
Facing metals	3.2.3.7	4.5.1	X		X	
Adjacent sensors	3.2.3.8	4.5.1	X		X	
Magnetic attraction	3.2.3.9	4.4.3	X			
Switch stabilization	3.2.3.10	4.5.1	X		X	
Weight	3.2.4.1	4.4.3	X			
Dimensions	3.2.4.2	4.4.3	X			
Cable	3.2.4.3	4.4.3	X			
Encapsulation	3.2.4.4	4.4.3	X			
Materials	3.3	4.4.3	X			
Reliability	3.2.5	4.5.2	X			

See footnotes at end of table.

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TABLE I. Examination and tests - Continued

	Requirement	Test method	First article	Quality conformance inspection		
				Group A	Group B	Group C
Electromagnetic compatibility	3.4	4.5.7	X			
Dielectric withstand voltage	3.2.1.7 and 3.2.2 7	4.5.8	X		X2/	
Insulation resistance	3.2 1.8 and 3.2.2 8	4 5.9	X	X2/		
Temperature	3 2 7 1	4.5 3	X			X
Humidity	3.2.7 2	4.5 3	X			
Vibration	3.2 7.3	4.5 4	X			
Shock	3.2.7 4	4.5.5	X			
Salt spray	3.2 7 5	4.5.6	X			
AFFF compatibility	3.2.7.6	4.5 10	X			

1/ Resistive load only
2/ Momentary measurement only

4.3 1 First article sample One switch of the designation submitted (see 1.2 1) shall be subjected to first article inspection

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2 1), quality conformance inspection shall consist of group A, group B, and group C inspections, as specified in table I.

4.4 1 Inspection lot An inspection lot shall consist of all switches of the same designation, produced under essentially the same conditions and offered for inspection at one time

4.4.2 Sampling. Group A sampling shall consist of every item produced Groups B and C shall consist of two parts from the first 50 parts produced. For subsequent group B and C inspection, two parts from each successive 100 parts, or two parts per month if 100 parts are produced monthly, shall be selected

4.4.3 Production inspection (group A). Unless otherwise specified (see 6.2.1), production inspection shall be made on every item offered for delivery. This inspection shall include the examination and tests which will assess the workmanship and reveal omissions and errors of the production process Examples of tests are. functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Unless otherwise specified (see 6 2 1), production inspection shall include the examination and tests shown in group A of table I and shall be performed as approved in accordance with 4.5

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4.4.4 Production control inspection (group B). Production control inspection shall be conducted on a sampling basis as specified (see 6.2.1). This inspection shall consist of the examination and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Unless otherwise specified (see 6.2.1), production control inspection shall include the examination and tests shown in group B, of table I, and shall be performed as approved in accordance with 4.5.

4.4.5 Environmental inspection (group C). Environmental inspection shall be conducted on a sampling basis as specified (see 6.2.1). This inspection shall encompass environmental tests to assess the durability of the materials, parts, units, and the equipment as a whole; life tests, simulated service tests; test of the effects of changes of environment (such as extremes of temperature and humidity, effect of salt air); and tests of the effect of shock, vibration and inclination. Unless otherwise specified (see 6.2.1), environmental inspection shall include the examination and tests shown in group C of table I, and perform as approved in accordance with 4.5.

4.5 Tests.

4.5.1 Operational test. Operational tests shall be performed to verify electrical and operational performance characteristics and shall meet the requirements specified herein.

4.5.2 Accelerated life test. Sample switch shall be subjected to 720 hours of continuously cycled operation in 70°C (158°F) ambient at full rated load on each output. The switching rate shall be actuation cycles per minute. Switches shall meet the requirements specified in 3.2.5.

4.5.3 Temperature and humidity tests. Temperature and humidity tests shall be conducted in accordance with MIL-STD-202 and shall meet the requirements specified in 3.2.7.1 and 3.2.7.2. Switches shall be continuously cycled throughout the test.

4.5.4 Vibration. The switch shall be subjected to the type I vibration test as specified in MIL-STD-167-1 and shall meet the requirements specified in 3.2.7.3. Circuits shall be energized during the test. Test equipment shall monitor the output of the switch. Oscillograph readings shall be taken during the last minute of the final tests along each principal axis. Test shall be repeated with target and without target present (that is, switch actuated and deactuated). Failure to conform to the requirements of 3.2.7.3 shall be cause for rejection.

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4 5.5 Shock. The switch shall be subjected to the high-impact shock test for grade A, type A, class I, lightweight equipment as specified in MIL-S-901 and shall meet the requirements specified in 3.2.7.4. Circuits shall be energized. Test equipment shall monitor output of the switch. Test shall be repeated with target and without target present (that is, switch actuated and deactuated). Failure to conform to the requirements of 3.2.7.4 shall be cause for rejection.

4.5.6 Salt spray. Switches shall be tested as specified in method 101, test condition A of MIL-STD-202 and shall meet the requirements specified in 3.2 7.5. Evidence of excessive corrosion that will cause a failure of the switch shall be cause for rejection.

4.5.7 Electromagnetic compatibility. Switches shall be tested in accordance with MIL-STD-462 In addition, normal operation shall be demonstrated in pulsed and continuous wave radio frequency fields throughout the frequency ranges of 2 to 32 MHz, 108 to 400 MHz and 2 to 18 gigahertz, with the attached cable exposed. Switches shall meet the requirements specified in 3 4

4.5 8 Dielectric withstanding voltage Switches shall be subjected to a dielectric withstanding voltage test as specified in method 301 of MIL-STD-202 The test voltage shall be plus or minus 250 Vdc. Points of application shall be between each individual cable conductor and case, but not between conductors. The switches shall meet the requirements specified in 3 2.1 7 and 3 2 2 7.

4 5 9 Insulation resistance Insulation resistance shall be measured as specified in method 302, test condition A of MIL-STD-202 for a duration of 2 minutes. The test voltage shall be applied between each cable conductor and case, but not between conductors. The switches shall meet the requirements specified in 3.2 1.8 and 3.2.2.8.

4 5 10 AFFF compatibility Switches shall be immersed to one-half of their length in a bath of AFFF for 96 continuous hours. The AFFF bath shall be a solution of 6 percent AFFF in a 5 percent concentration of salt water The solution and the ambient temperature shall be 35°C (95°F) The solution shall be kept agitated enough to keep the unimmersed part of the switch covered with foam The switches shall meet the requirements specified in 3 2.7 6 After immersion, switches shall be placed in an environment of 35°C (95°F) and 95 percent relative humidity for 25 days and cycled a minimum of once per minute If the switch does not open and close for any of the cycles, it shall constitute a failure. The switch shall then be disassembled and examined for internal salt deposit or corrosion Either condition shall constitute a failure.

4.6 Inspection of packaging. Sample packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Packaging requirements. Switches shall be preserved and packed level A, or C, as specified (see 6.2.1) and marked in accordance with MIL-S-28786 and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2.1). In addition, for Navy acquisitions, the following applies:

(a) Navy fire-retardant requirements.

(1) Lumber and plywood. Unless otherwise specified (see 6.2 1), all lumber and plywood including laminated veneer material used in shipping container construction members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

(2) Fiberboard. Unless otherwise specified (see 6 2 1), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements of PPP-F-320 and amendments thereto.

6. NOTES

6.1 Intended use. The proximity switches are for use in position sensing applications.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Class and designation of switches required (see 1.2 and 1.2.1)
(c) When first article inspection is required (see 3.1).
(d) If quality conformance inspection is other than specified (see 4.4).
(e) Sampling basis for production inspection (group A) if other than specified (see 4.4.3).
(f) The examinations and tests for production inspection (group A) if other than specified (see 4.4.3).

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(g) Sampling basis for production control inspection (group B) (see 4.4.4).
(h) The examinations and tests for production control inspection (group B) if other than specified (see 4.4.4).
(i) Sampling basis for environmental inspection (group C) (see 4.4.5).
(j) The examinations and tests for environmental inspection (group C) if other than specified (see 4.4.5).
(k) Level of preservation and packing required (see 5.1)
(l) When fire-retardant materials are not required (see 5.1).
(m) Marking requirements (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements Deliverable data required by this specification is cited in the following paragraph

Paragraph no.	Data requirement title	Applicable DID no.	Option
3 6	Drawings, engineering and associated lists	DI-E-7031	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2 2 1 The data requirements of 6.2 2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Definitions. Terms used in this specification are defined in 6 4 1 through 6.4.16 and graphically shown on figure 1.

6.4.1 Sensing face. The sensing face is a surface of the proximity switch or photoelectronic sensor reacting to an alternating electromagnetic field or radiant energy source.

6.4.2 Reference axis. The reference axis is perpendicular to the sensing face and passes through its center.

6.4.3 Target. Target refers to the item, material, or configuration that a sensor is intended to detect.

6.4.4 Mode of operation. Mode of operation is the direction of actuator travel in relationship to the sensing face of the sensor.

6.4.5 Head-on mode. Head-on mode indicates when the actuator is parallel and centered on the sensitive face of the sensor, and its path of travel is perpendicular to the sensitive face along the reference axis. The performance criteria are given as dimensions from the sensing face to the leading edge of the actuator.

6.4.6 Slide-by mode. Slide-by mode indicates when the actuator and its path of travel are parallel to the sensing face of the sensor The performance criteria are given as dimensions from the reference axis to the leading edge of the actuator.

6.4.7 Actuation point. The actuation point is the point at which the target approaching the sensing face is detected.

6.4.8 Deactuation point. The deactuation point is the point at which the target leaving the sensing zone is no longer detected.

6.4.9 Hysteresis. Hysteresis is the distance between the actuation point and the deactuation point measured in the direction of target travel In the head-on mode, it is measured along the reference axis. In the slide-by mode, it is measured in a plane parallel to the sensing face along a line perpendicular to the reference axis passing through the reference axis in the direction of target travel.

6.4.10 Sensing distance. Sensing distance as the perpendicular distance from the actuation point to the sensing face in the head-on mode. It is the perpendicular distance from the actuation point to the reference axis in the slide-by mode.

6.4.11 Repeatability. Repeatability is the maximum shift in the sensing distance when the switch is operated anywhere within specified parameter limits

6.4.12 Source. Source is the device which switches a positive voltage to a load. The current flows through the sensor and into the load

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6.4.13 Sink. Sink is a device which switches the load to ground The current flows through the load and into the sensor before being switched to ground.

6.4.14 Complementary proximity sensor. A complementary proximity sensor features a normally open and a normally closed output, both of which can be used simultaneously.

6.4.15 Make current rating. Make current rating is the maximum surge current a device will conduct when energizing a load.

6.4.16 Break current rating. Break current rating is the steady-state current flowing in a device at the time current is interrupted

6 5 Application guidance notes.

6 5.1 Cable recommendations Omniguard 2, Dekoron TPE on Plenum cable, or Belden Plenum cables have demonstrated physical properties appropriate to this specification. Other cables, including NSN 6145-00-110-2262, may be equally appropriate.

6 5 2 Potting compound recommendations Rigid epoxy in accordance with MIL-I-16923, self-extinguishing, is an example of a suitable compound for encapsulation.

6.5.3 Casing material recommendations Stainless steel (316) has attractive chemical and physical properties for the switch casing mounting plate, if used

6.6 Sub-contracted material and parts The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6 7 Subject term (key word) listing

Aqueous film forming foam
Head-on mode
Hysteresis
Inductive sensor
Sensing face

Preparing activity:
Navy - SH
(Project 5930-N625)

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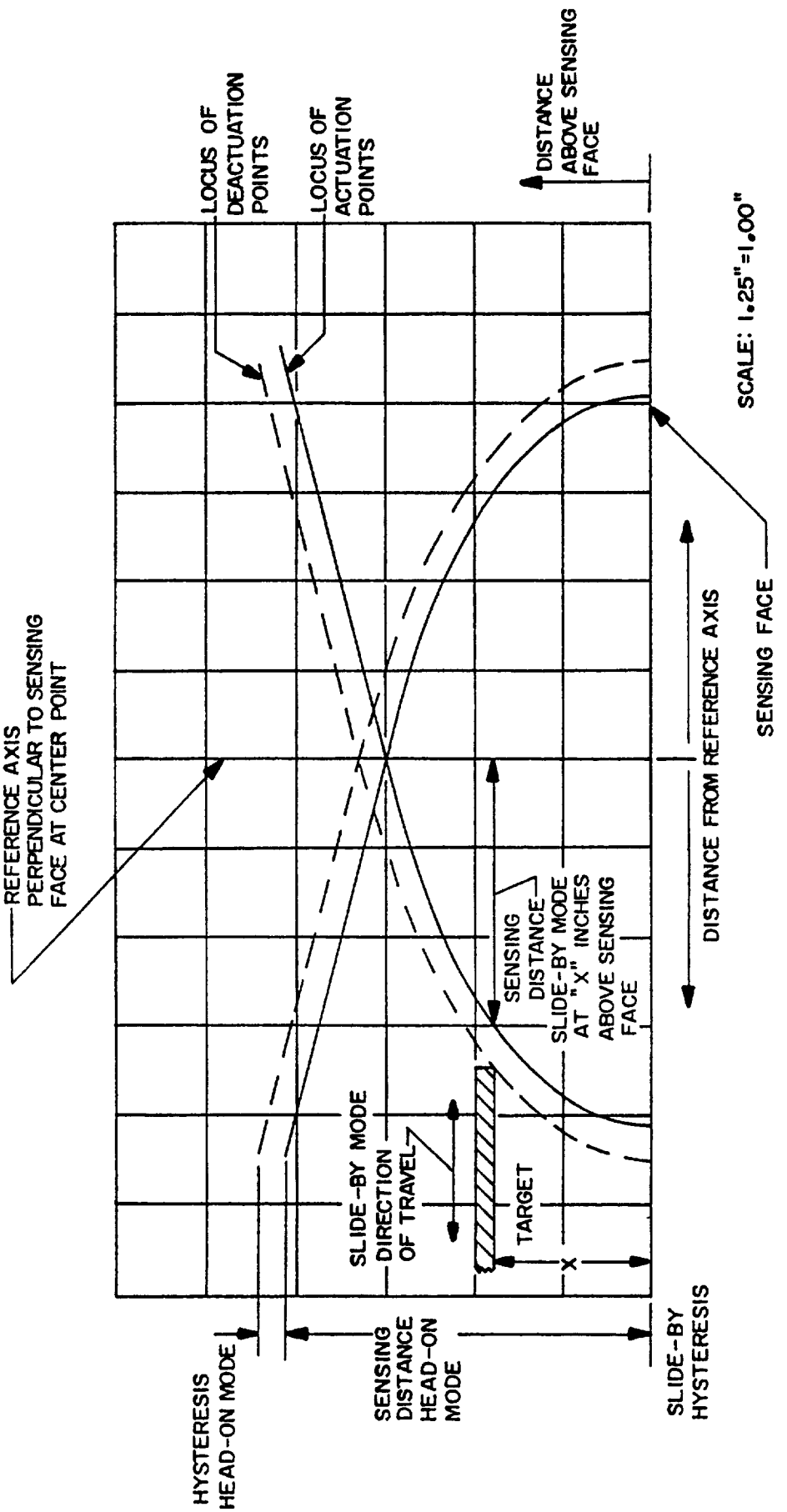


FIGURE 1 Actuation curve with hysteresis.

SH 13203004

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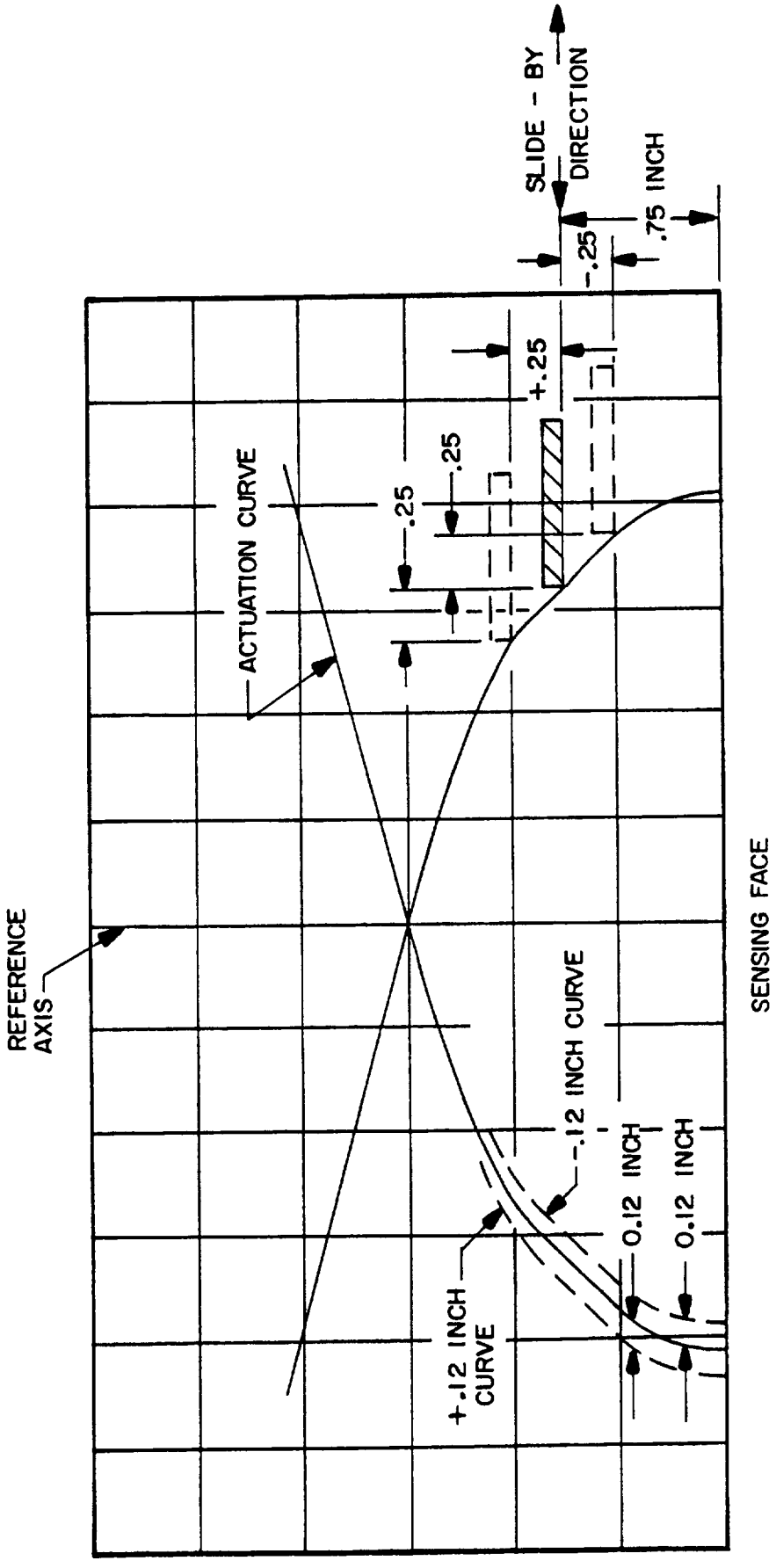


FIGURE 2. Repeatability curve slide-by mode, 1.5-inch switch.

SH 1320300S

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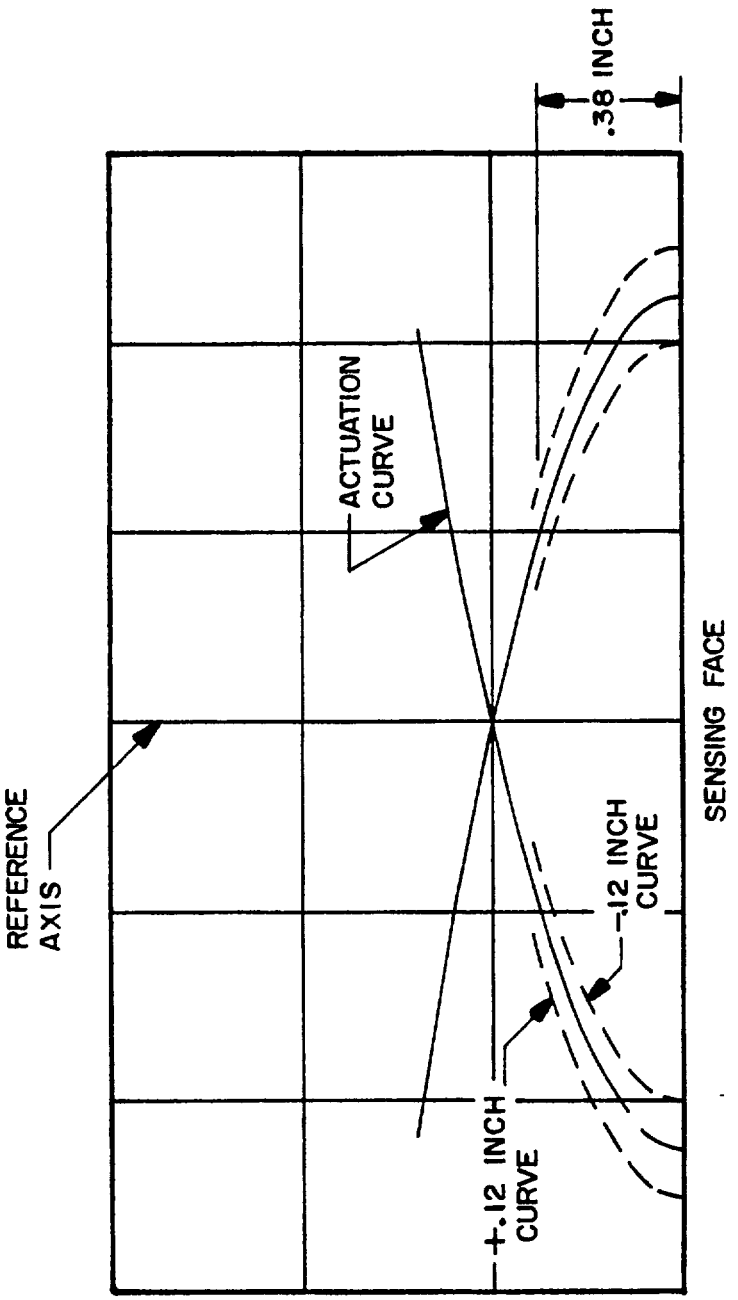
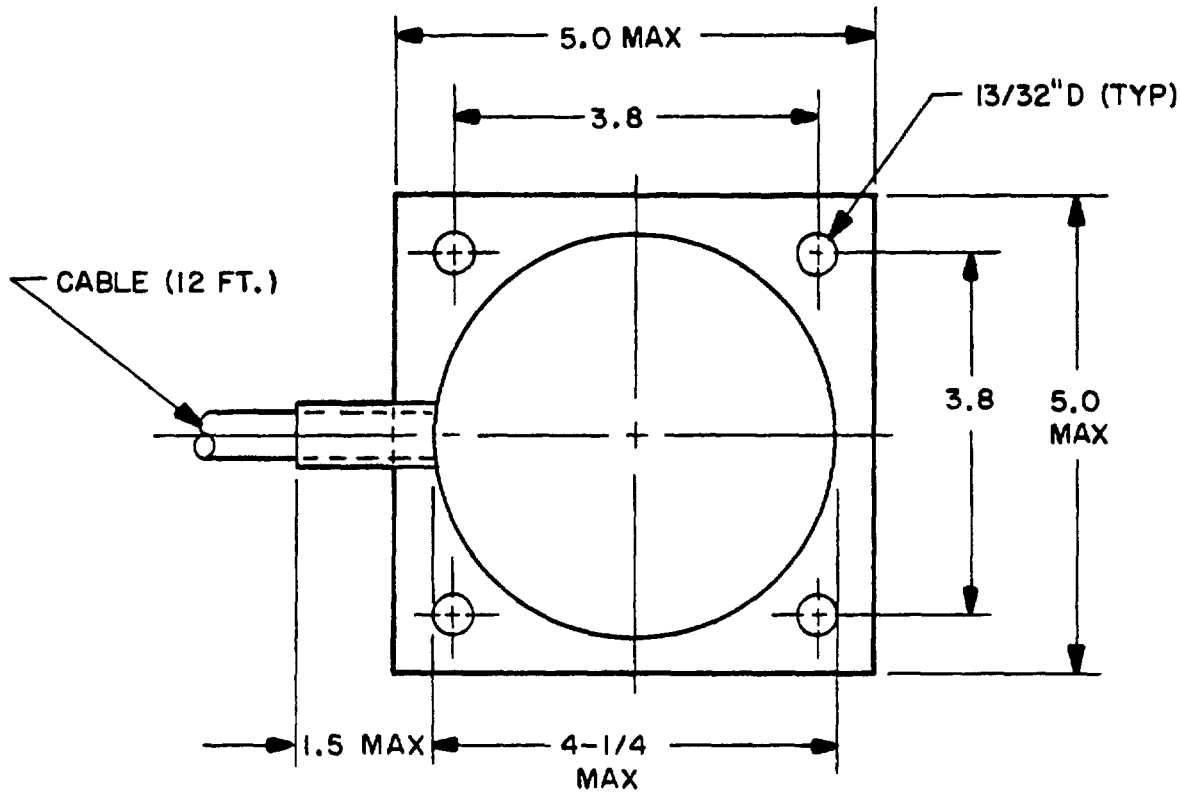
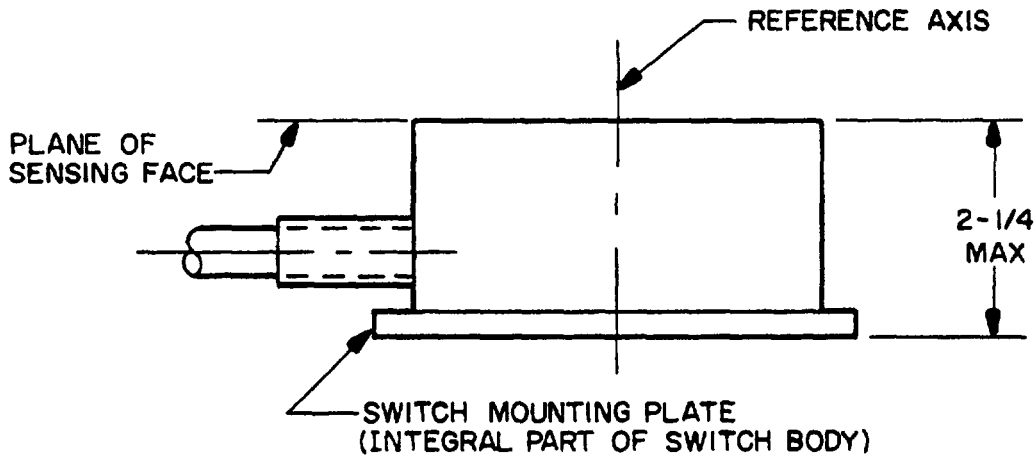


FIGURE 3. Repeatability curve slide-by mode, 0 5-inch switch

SH 1203006

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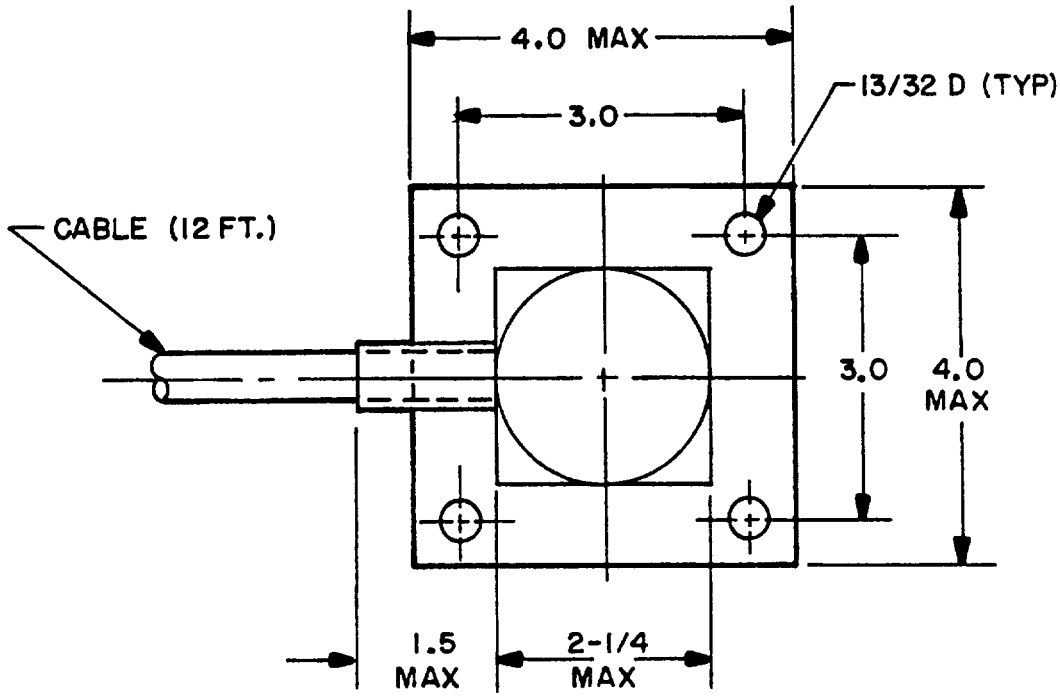


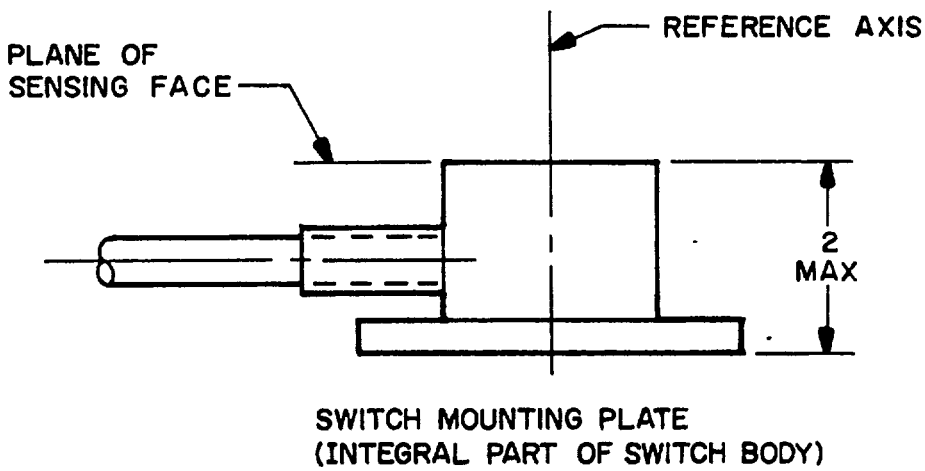


SH 13203007

FIGURE 4 Outline dimensions for dc - 1 5-NONC, ac-1.5-NO, or ac-1.5-NC switch

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SH 13203008

FIGURE 5 Outline dimensions for dc-0.5-NONC, ac-0.5-NO, or ac-0.5-NC switch

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-24711(SH)	SWITCH, PROXIMITY, SOLID-STATE

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426

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